Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Drug development with novel mechanisms of action to overcome resistance to targeted and immune-oncology therapies in pancreatic cancer			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Protein therapeutics	Medical Bio	Small molecule pharmaceuticals
Concept of Proposed Technology	This proposal focuses on the discovery and preclinical validation of **peptide or small molecule anticancer drug candidates** designed to address **novel resistance mechanisms** and/or **tumor-selective vulnerabilities** relevant to **pancreatic cancer**. The candidates addressed under this proposal have progressed to at least **Technology Readiness Level (TRL) 3**, having demonstrated preliminary mechanism-of-action validation and proof-of-concept efficacy in relevant *in vitro* and early *in vivo* models. These assets are positioned for preclinical advancement and translational development through international joint R&D with experienced academic research institutions and medical centers specializing in pancreatic cancer research. The technology emphasizes: • Targeting of resistance pathways that limit the efficacy and durability of current therapies in pancreatic cancer			

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	• Utilization of peptide and small-molecule modalities as alternatives or complements to existing biologics, offering enhanced tumor penetration and bioavailability • Integration of translational biomarker strategies to support rational patient selection and response assessment • Development of novel pancreatic cancer efficacy models to better recapitulate patient-relevant resistance mechanisms
Rationale for Joint R&D Project	The purpose of this joint R&D framework is to establish a **collaborative platform** through which bio-ventures that have achieved **TRL 3** can partner with international academic research institutions experienced in translational oncology and pancreatic cancer research. By leveraging complementary strengths in **early-stage drug development, novel efficacy model development, *in vivo* pharmacology, and preclinical-to-clinic translation,** the project aims to advance qualified candidates toward **TRL 4**. The joint program emphasizes: • Advanced *in vivo* efficacy testing in physiologically relevant pancreatic cancer models • Development of novel pancreatic cancer efficacy models incorporating resistance mechanisms • Biomarker discovery and validation strategies aligned with pancreatic cancer biology • Preparation for subsequent preclinical regulatory pathways and clinical development planning
Key Words	1. Peptide/small-molecule targeted therapy for pancreatic cancer 2. Pancreatic cancer resistance mechanisms and tumor-specific vulnerabilities 3. Novel efficacy model development for pancreatic cancer 4. *In vivo* pharmacology and translational biomarker strategies 5. TRL 3-4 preclinical and translational development

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (4)
R&D Objectives	**R&D Objectives** • To support advancement of peptide or small-molecule anticancer candidates for **pancreatic cancer** from **TRL 3 to TRL 4** through joint R&D between bio-ventures and international academic research institutions • To develop **first-in-class or best-in-class therapies** targeting resistance mechanisms in pancreatic cancer • To establish a **translational ecosystem** integrating pancreatic cancer resistance biology, novel efficacy models, biomarker strategies, and mechanistic studies to enable rational progression toward IND-enabling studies • To develop and validate novel in vivo pancreatic cancer models that better recapitulate patient-relevant resistance mechanisms and therapeutic response			

R&D Target Specifications	**TRL 3 Entry Criteria**
	
	• At least one peptide or small-molecule candidate demonstrating:
	  ○ Proven mechanism of action in pancreatic cancer relevant cell lines or tissues
	  ○ Preliminary efficacy signals in early in vivo models of pancreatic cancer
	  ○ Basic characterization of chemical/physical properties and manufacturability
	  ○ Defined target engagement and pathway modulation relevant to pancreatic cancer resistance
	
	TRL 4 Targets During the Project
	
	• **Reproducible in vivo antitumor efficacy** in pancreatic cancer models, supported by appropriate study design
	• Development and validation of at least one new(or substantially enhanced) pancreatic cancer in vivo efficacy model, including:
	  ○ Model establishment and characterization
	  ○ Demonstration of suitability for detecting treatment effects
	  ○ Standard operating procedures
	• Determination of dose-responsive antitumor activity in in vivo pancreatic cancer models
	• Characterization of pharmacokinetic properties and proof -of-mechanism biomarkers in in vivo studies
	• Initial in vivo safety and tolerability assessment in rodent models
	• Generation of a comprehensive preclinical data package supporting IND/CTA-enabling studies
	• Preliminary assessment of combination potential with standard-of-care pancreatic cancer therapies			

R&D Plan	• **Candidate and Hypothesis Consolidation**
	  ○ Confirm target hypothesis, differentiation rationale, and Go/No-Go criteria for TRL 4 gate
	  ○ Finalize candidate selection and material specifications for in vivo testing
	
	• **In Vivo Efficacy and Mechanism Support**
	  ○ Generate supporting in vitro activity data in pancreatic cancer-relevant assays
	  ○ Define PD marker candidates and sampling strategy to enable mechanism-supportive readouts in vivo.
	• **In Vivo Efficacy**
	  ○ Conduct dose-ranging and schedule-finding pilot studies
	  ○ Execute confirmatory efficacy studies with predefined endpoints			

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	• **New in Vivo Efficacy Model Development**
	  ○ Develop and validate new pancreatic cancer efficacy models aligned with the project's resistance biology focus
	  ○ Establish model quality parameters and SOPs to support reproducibility and future expansion
	• **Data Package and TRL 4 Gate Review**
	  ○ Integrate efficacy, exposure, and PD evidence into a TRL 4 POC package
	  ○ Summarize risks, limitations, and a follow-on plan for TRL 5 progression

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead *in vivo* efficacy study design and execution in pancreatic cancer, including dosing strategy, controls, endpoints, and statistical considerations appropriate for TRL 4
	• Lead development and validation of new pancreatic cancer *in vivo* efficacy models, including:
	  ○ Model establishment, characterization, and documentation(SOP)
	  ○ Performance benchmarking(variability, dynamic range, suitability for detecting treatment effects)
	• Conduct preliminary *in vivo* pharmacology studies, including:
	  ○ Dose-dependancy studies in pancreatic cancer models
	  ○ Basic validation of sufficient pharmacokinetic and/or pharmacodynamic properties
	  ○ Mechanism-of-action and biomarker analysis
	  ○ Gross safety and tolerability assessment
	• Assess combination potential with standard pancreatic cancer therapies
	• Explore potential predictive biomarkers supporting rational patient selection
	• Provide expert input on pancreatic cancer biology, translational strategy, and advancement to IND/CTA-enabling studies

Expected R&R of Korean SME Partner	• Provide peptide or small-molecule candidates that have achieved at least TRL 3, including supporting preclinical efficacy and mechanism-of-action data • Supply sufficient quantities of investigational materials and coordinate shipment/logistics; manage batch documentation and quality checks appropriate for research use • Support assay development and provide mechanistic context; collaborate on defining PD markers and study hypothesis • Lead overall project management from the Korean side, including budget control, reporting, coordination with TIPA requirements, and integrated documentation of deliverables. • Define a follow-on development roadmap toward TRL 5(e.g., early CMC considerations and nonclinical planning), without committing to IND-enabling scope within this TRL 3-4 project

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** • Establishment of novel *in vivo* pancreatic cancer efficacy models that better recapitulate patient-relevant resistance mechanisms and serve as valuable platform assets for future drug discovery • Preclinical validation of peptide or small-molecule therapeutics targeting pancreatic cancer resistance pathways, progressed from TRL 3 to TRL 4 • Development of a robust translational framework linking pancreatic cancer biology, resistance mechanisms, biomarkers, and drug efficacy • Predictive biomarker signatures supporting rational patient selection for subsequent clinical trials • Comprehensive preclinical safety and efficacy data enabling advancement toward IND/CTA filing
	Scientific and Clinical Impact • Generation of high-quality preclinical evidence in physiologically relevant pancreatic cancer models, addressing a major unmet medical need • Novel mechanistic insights into pancreatic cancer resistance and therapeutic vulnerabilities • Enhancement of translational research capacity in pancreatic cancer drug development • Acceleration of progression toward IND-enabling studies and early-phase clinical trials • Strengthening of Korea's role in resistance-focused oncology and translational pancreatic cancer research
	Industrial and Economic Impact • Enhancement of Korean bio-venture global competitiveness through co-development of differentiated pancreatic cancer therapeutics • Expansion of licensing-out and partnering opportunities with international pharmaceutical companies • Establishment of long-term research partnerships with leading international academic institutions • Contribution to the domestic biopharmaceutical ecosystem through technology advancement, workforce development, and international collaboration • Potential for follow-on funding from international sources based on strong preclinical data • Economic impact from future commercialization of pancreatic cancer therapeutics

6. References

(1) Peptide inhibitors: Breaking cancer code, Eur. J. Med. Chem. 297 (2025) 117961

(2) Cancer testis antigens: Emerging therapeutic targets leveraging genomic instability in cancer, Mol. Ther. Oncol. 32 (2024) 200768

(3) Advance in peptide-based drug development: delivery strategies and clinical progress, Signal Transduct Target Ther. 10: 74 (2025)

(4) *In vivo* models of pancreatic ductal adenocarcinoma, Cancer Metastasis Rev. 42(1):1-24(2023)